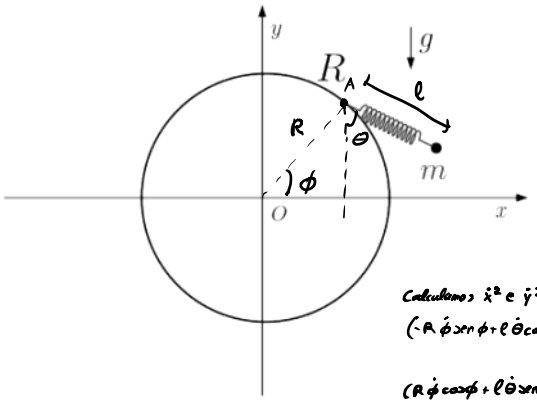
$N = 1$

$f = 3$



Definimos las posiciones de los siguientes puntos

A: $x_A = R\cos\phi$
$y_A = R\,\mathrm{sen}\,\phi$

m: $x_m = x_A + \ell\,\mathrm{sen}\,\theta$
$y_m = y_A - \ell\cos\theta$

Calculamos las velocidades correspondientes:

$\dot{x}_m = -R\dot{\phi}\,\mathrm{sen}\,\phi + \ell\dot{\theta}\cos\theta + \dot{\ell}\,\mathrm{sen}\,\theta$
$\dot{y}_m = R\dot{\phi}\cos\phi + \ell\dot{\theta}\,\mathrm{sen}\,\theta - \dot{\ell}\cos\theta$

Calculamos la energía cinética y potencial de la masa m:

$E_c = \frac{1}{2}mv^2 = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2)$

Calculamos $\dot{x}^2$ e $\dot{y}^2$:

$(-R\dot{\phi}\,\mathrm{sen}\,\phi + \ell\dot{\theta}\cos\theta + \dot{\ell}\,\mathrm{sen}\,\theta)^2 = R^2\dot{\phi}^2\,\mathrm{sen}^2\phi + \ell^2\dot{\theta}^2\cos^2\theta + \dot{\ell}^2\,\mathrm{sen}^2\theta + 2(-R\dot{\phi}\ell\dot{\theta}\,\mathrm{sen}\,\phi\cos\theta + R\dot{\phi}\dot{\ell}\,\mathrm{sen}\,\phi\,\mathrm{sen}\,\theta - \ell\dot{\ell}\dot{\theta}\cos\theta\,\mathrm{sen}\,\theta) = \dot{x}^2$

$(R\dot{\phi}\cos\phi + \ell\dot{\theta}\,\mathrm{sen}\,\theta - \dot{\ell}\cos\theta)^2 = R^2\dot{\phi}^2\cos^2\phi + \ell^2\dot{\theta}^2\,\mathrm{sen}^2\theta + \dot{\ell}^2\cos^2\theta + 2(R\ell\dot{\phi}\dot{\theta}\cos\phi\,\mathrm{sen}\,\theta - R\dot{\ell}\dot{\phi}\cos\phi\cos\theta - \ell\dot{\ell}\dot{\theta}\,\mathrm{sen}\,\theta\cos\theta) = \dot{y}^2$

c)

La suma de ambas quedaría entonces de la siguiente manera: (Utilizamos las propiedades de suma y resta de identidades trigonométricas: $\begin{cases} \mathrm{sen}^2\alpha + \cos^2\alpha = 1 \\ \cos(\phi-\theta) = \cos\phi\cos\theta - \mathrm{sen}\,\phi\,\mathrm{sen}\,\theta \\ \mathrm{sen}(\phi-\theta) = \mathrm{sen}\,\phi\cos\theta - \mathrm{sen}\,\theta\cos\theta \end{cases}$)

$v^2 = R^2\dot{\phi}^2 + \dot{\ell}^2 + \ell^2\dot{\theta}^2 - 2R\dot{\phi}\dot{\ell}\cos(\phi-\theta) + 2R\ell\dot{\phi}\dot{\theta}\,\mathrm{sen}(\theta-\phi)$

Por lo tanto, la expresión de la energía cinética de la partícula m es:

$E_c = \frac{1}{2}m(R^2\dot{\phi}^2 + \dot{\ell}^2 + \ell^2\dot{\theta}^2 - 2R\dot{\phi}\dot{\ell}\cos(\phi-\theta) + 2R\ell\dot{\phi}\dot{\theta}\,\mathrm{sen}(\theta-\phi))$

Para calcular la energía potencial debemos calcular la potencial gravitatoria y la del muelle, dado que son las únicas fuerzas conservativas que actúan en el sistema:

$V_g = mgy_m = mg(R\,\mathrm{sen}\,\phi - \ell\cos\theta)$

Para la del muelle, sabemos que $F = K\Delta\ell = K(\ell-\ell_0) \Rightarrow V_e = \int_{\ell_0}^{\ell} K(s-\ell_0)\,ds$ → Hacemos el siguiente cambio de variable: $\begin{cases} s-\ell_0 = x \\ dx = ds \end{cases} \Rightarrow V_e = \int_{\ell_0}^{\ell} Kx\,dx = \frac{1}{2}Kx^2\Big|_{\ell_0}^{\ell} = \frac{1}{2}K(s-\ell_0)^2\Big|_{\ell_0}^{\ell} \Rightarrow V_e = \frac{1}{2}K(\ell-\ell_0)^2$

$V_e = \frac{1}{2}K(\ell-\ell_0)^2$

El potencial entonces es:

$V = V_g + V_e = mg(R\,\mathrm{sen}\,\phi - \ell\cos\theta) + \frac{1}{2}K(\ell-\ell_0)^2$

Formamos la lagrangiana

$\mathcal{L} = \frac{1}{2}m(R^2\dot{\phi}^2 + \dot{\ell}^2 + \ell^2\dot{\theta}^2 - 2R\dot{\phi}\dot{\ell}\cos(\phi-\theta) + 2R\ell\dot{\phi}\dot{\theta}\,\mathrm{sen}(\theta-\phi)) - mg(R\,\mathrm{sen}\,\phi - \ell\cos\theta) - \frac{1}{2}K(\ell-\ell_0)^2$

Según la ecuación de Euler-Lagrange:

$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}_j}\right) - \frac{\partial\mathcal{L}}{\partial q_j} = 0 \qquad j = 1,2,3 \begin{cases} q_1 = \phi \\ q_2 = \theta \\ q_3 = \ell \end{cases}$

- ϕ)

$\frac{\partial\mathcal{L}}{\partial\phi} = mR(\dot{\phi}\dot{\ell}\,\mathrm{sen}(\phi-\theta) - \ell\dot{\theta}\cos(\phi-\theta)) - mgR\cos\phi$

$\frac{\partial\mathcal{L}}{\partial\dot{\phi}} = mR(R\dot{\phi} - \dot{\ell}\cos(\phi-\theta) - \ell\dot{\theta}\,\mathrm{sen}(\phi-\theta))$

$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\phi}}\right) = mR(R\ddot{\phi} - \ddot{\ell}\cos(\phi-\theta) + \dot{\ell}(\dot{\phi}-\dot{\theta})\,\mathrm{sen}(\phi-\theta) - \dot{\ell}\dot{\theta}\,\mathrm{sen}(\phi-\theta) - \ell\ddot{\theta}\,\mathrm{sen}(\phi-\theta) - \ell\dot{\theta}(\dot{\phi}-\dot{\theta})\cos(\phi-\theta))$

Por ello:

$R\ddot{\phi} - \ddot{\ell}\cos(\phi-\theta) + \dot{\ell}(\dot{\phi}-\dot{\theta})\,\mathrm{sen}(\phi-\theta) - \dot{\ell}\dot{\theta}\,\mathrm{sen}(\phi-\theta) - \ell\ddot{\theta}\,\mathrm{sen}(\phi-\theta) - \ell\dot{\theta}(\dot{\phi}-\dot{\theta})\cos(\phi-\theta) - R\dot{\phi}\dot{\ell}\,\mathrm{sen}(\phi-\theta) + \ell\dot{\phi}\cos(\phi-\theta) + gR\cos\phi = 0$

- θ)

$\frac{\partial\mathcal{L}}{\partial\theta} = m(-R\dot{\phi}\dot{\ell}\,\mathrm{sen}(\phi-\theta) + R\ell\dot{\phi}\dot{\theta}\cos(\phi-\theta)) - mgl\,\mathrm{sen}\,\theta$

$\frac{\partial\mathcal{L}}{\partial\dot{\theta}} = m(\ell^2\dot{\theta} - R\ell\dot{\phi}\,\mathrm{sen}(\phi-\theta))$

$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right) = m(\ell^2\ddot{\theta} + 2\ell\dot{\ell}\dot{\theta} - R\ell\dot{\phi}\,\mathrm{sen}(\phi-\theta) - R\ell\ddot{\phi}\,\mathrm{sen}(\phi-\theta) - R\ell\dot{\phi}(\dot{\phi}-\dot{\theta})\cos(\phi-\theta))$

Por ello:

$\ell\ddot{\theta} + 2\dot{\ell}\dot{\theta} - R\ddot{\phi}\,\mathrm{sen}(\phi-\theta) - R\dot{\phi}^2\cos(\phi-\theta) + g\,\mathrm{sen}\,\theta = 0$

- ℓ)

$\frac{\partial\mathcal{L}}{\partial\ell} = m(\ell\dot{\theta}^2 - R\dot{\phi}\dot{\theta}\,\mathrm{sen}(\phi-\theta)) + mg\cos\theta - K(\ell-\ell_0)$

$\frac{\partial\mathcal{L}}{\partial\dot{\ell}} = m(\dot{\ell} - R\dot{\phi}\cos(\phi-\theta))$

$\frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{\ell}}\right) = m(\ddot{\ell} - R\ddot{\phi}\cos(\phi-\theta) + R\dot{\phi}(\dot{\phi}-\dot{\theta})\,\mathrm{sen}(\phi-\theta))$

Por ello:

$m\ddot{\ell} - m\ell\dot{\theta}^2 - mR\ddot{\phi}\cos(\phi-\theta) + mR\dot{\phi}^2\,\mathrm{sen}(\phi-\theta) - mg\cos\theta + K(\ell-\ell_0) = 0$

El sistema de ecuaciones quedaría así:

$$\begin{cases} R\ddot{\phi} - \ddot{\ell}\cos(\phi-\theta) + \dot{\ell}(\dot{\phi}-\dot{\theta})\,\mathrm{sen}(\phi-\theta) - \ell\dot{\theta}\,\mathrm{sen}(\phi-\theta) - \ell\ddot{\theta}\,\mathrm{sen}(\phi-\theta) - \ell\dot{\theta}(\dot{\phi}-\dot{\theta})\cos(\phi-\theta) - R\dot{\phi}\dot{\ell}\,\mathrm{sen}(\phi-\theta) + \ell\dot{\phi}\cos(\phi-\theta) + gR\cos\phi = 0 \\ \ell\ddot{\theta} + 2\dot{\ell}\dot{\theta} - R\ddot{\phi}\,\mathrm{sen}(\phi-\theta) - R\dot{\phi}^2\cos(\phi-\theta) + g\,\mathrm{sen}\,\theta = 0 \\ m\ddot{\ell} - m\ell\dot{\theta}^2 - mR\ddot{\phi}\cos(\phi-\theta) + mR\dot{\phi}^2\,\mathrm{sen}(\phi-\theta) - mg\cos\theta + K(\ell-\ell_0) = 0 \end{cases}$$

Para determinar las constantes de movimiento analizamos qué coordenadas generalizadas son cíclicas (no aparecen explícitamente en la lagrangiana).
Como podemos observar, las tres coordenadas (ϕ, θ, ℓ) aparecen explícitamente en la expresión de $\mathcal{L}$, por lo que ninguno de los momentos conjugados asociados a ellas ($\pi_j = \frac{\partial \mathcal{L}}{\partial \dot{q}_j}$, $j=1,2,3$) es una constante del movimiento.

Sin embargo, la lagrangiana no depende explícitamente del tiempo $\left(\frac{\partial \mathcal{L}}{\partial t} = 0\right)$, por lo que la función $\tilde{H}$ definida por:

$$\tilde{H} = \sum_{j=1}^{f} \dot{q}_j \frac{\partial \mathcal{L}}{\partial \dot{q}_j} - \mathcal{L} = cte., \ f=3, \text{ se conserva, por lo que es una constante de movimiento.}$$

Además, como V no depende explícitamente de las velocidades generalizadas $\dot{q}_j$ y las ligaduras a las que se encuentra sujeto el sistema son esclerónomas (no dependen del tiempo), la función $\tilde{H}$ coincide con la energía mecánica del sistema $E = T+V$. Por ello:

$$E = \frac{1}{2}m\left(R^2\dot{\phi}^2 + \dot{\ell}^2 + \ell^2\dot{\theta}^2 - 2R\ell\dot{\phi}\dot{\ell}\cos(\phi-\theta) - 2R\ell\dot{\phi}\dot{\theta}\operatorname{sen}(\phi-\theta)\right) + mg\left(R\operatorname{sen}\phi - \ell\cos\theta + \frac{1}{2}k(\ell-\ell_0)^2 = cte.\right.$$

d)

Creamos una variable generalizada $r(t)$ que mida la distancia hacia el centro del disco O: $f(r) = r - R = 0$. Calculamos las coordenadas generalizadas en función de la nueva variable r:

$x_m = r\cos\phi + \ell\operatorname{sen}\theta$
$y_m = r\operatorname{sen}\phi - \ell\cos\theta$ $\Rightarrow$ Suponemos primero que $\ddot{r} \neq 0 \Rightarrow$ $\dot{x}_m = \dot{r}\cos\phi - r\dot{\phi}\operatorname{sen}\phi + \dot{\ell}\operatorname{sen}\theta + \ell\dot{\theta}\cos\theta$
$\dot{y}_m = \dot{r}\operatorname{sen}\phi + r\dot{\phi}\cos\phi - \dot{\ell}\cos\theta + \ell\dot{\theta}\operatorname{sen}\theta$

La velocidad elevada al cuadrado v^2 es: $v^2 = \dot{r}^2 + r^2\dot{\phi}^2 + \dot{\ell}^2 + \ell^2\dot{\theta}^2 - 2\dot{r}\dot{\ell}\operatorname{sen}(\phi-\theta) + 2\ell\dot{r}\dot{\theta}\cos(\phi-\theta) - 2r\dot{\phi}\dot{\ell}\cos(\phi-\theta) - 2r\ell\dot{\phi}\dot{\theta}\operatorname{sen}(\phi-\theta)$

La energía cinética T^* es entonces $T^* = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2 + \dot{\ell}^2 + \ell^2\dot{\theta}^2 - 2\dot{r}\dot{\ell}\operatorname{sen}(\phi-\theta) + 2\ell\dot{r}\dot{\theta}\cos(\phi-\theta) - 2r\dot{\phi}\dot{\ell}\cos(\phi-\theta) - 2r\ell\dot{\phi}\dot{\theta}\operatorname{sen}(\phi-\theta)\right)$

El potencial V^* sería entonces $V^* = mg(r\operatorname{sen}\phi - \ell\cos\theta) + \frac{1}{2}k(\ell-\ell_0)^2$

$$\mathcal{L}^* = T^* - V^* = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2 + \dot{\ell}^2 + \ell^2\dot{\theta}^2 - 2\dot{r}\dot{\ell}\operatorname{sen}(\phi-\theta) + 2\ell\dot{r}\dot{\theta}\cos(\phi-\theta) - 2r\dot{\phi}\dot{\ell}\cos(\phi-\theta) - 2r\ell\dot{\phi}\dot{\theta}\operatorname{sen}(\phi-\theta)\right) - mg(r\operatorname{sen}\phi - \ell\cos\theta) - \frac{1}{2}k(\ell-\ell_0)^2$$

Definimos la lagrangiana auxiliar $\mathcal{L}'$ de la siguiente manera:

$$\mathcal{L}' = \mathcal{L}^* + \lambda f(r) \Rightarrow \mathcal{L}' = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2 + \dot{\ell}^2 + \ell^2\dot{\theta}^2 - 2\dot{r}\dot{\ell}\operatorname{sen}(\phi-\theta) + 2\ell\dot{r}\dot{\theta}\cos(\phi-\theta) - 2r\dot{\phi}\dot{\ell}\cos(\phi-\theta) - 2r\ell\dot{\phi}\dot{\theta}\operatorname{sen}(\phi-\theta)\right) - mg(r\operatorname{sen}\phi - \ell\cos\theta) - \frac{1}{2}k(\ell-\ell_0)^2 - \lambda(r-R)$$

Asumimos ahora que la variable r es constante y es igual al radio del disco $\Rightarrow r = R, \dot{r} = 0, \ddot{r} = 0$, lo que nos permite calcular las fuerzas generalizadas de ligadura Q_i^L (en este caso hay una única fuerza de ligadura Q_r^L) de la siguiente forma:

$$Q_r^L = \frac{d}{dt}\left(\frac{\partial \mathcal{L}'}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i}, \ r = R, \dot{r} = 0, \ddot{r} = 0$$

$$\left.\frac{\partial \mathcal{L}'}{\partial r}\right|_{r=R} = m\left(R\dot{\phi}^2 - \dot{\phi}\dot{\ell}\cos(\phi-\theta) - \ell\dot{\phi}\dot{\theta}\operatorname{sen}(\phi-\theta)\right) - mg\operatorname{sen}\phi$$

$$\frac{\partial \mathcal{L}'}{\partial \dot{r}} = m\left(\dot{r} - \dot{\ell}\operatorname{sen}(\phi-\theta) + \ell\dot{\theta}\cos(\phi-\theta)\right)$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}'}{\partial \dot{r}}\right) = m\left(\ddot{r} - \ddot{\ell}\operatorname{sen}(\phi-\theta) - \dot{\ell}(\dot{\phi}-\dot{\theta})\cos(\phi-\theta) + \dot{\ell}\dot{\theta}\cos(\phi-\theta) + \ell\ddot{\theta}\cos(\phi-\theta) - \ell\dot{\theta}(\dot{\phi}-\dot{\theta})\operatorname{sen}(\phi-\theta)\right)$$

$$\left.\frac{d}{dt}\left(\frac{\partial \mathcal{L}'}{\partial \dot{r}}\right)\right|_{\substack{\dot{r}=0 \\ \ddot{r}=0}} = m\left(-\ddot{\ell}\operatorname{sen}(\phi-\theta) - \dot{\ell}(\dot{\phi}-\dot{\theta})\cos(\phi-\theta) + \dot{\ell}\dot{\theta}\cos(\phi-\theta) + \ell\ddot{\theta}\cos(\phi-\theta) - \ell\dot{\theta}(\dot{\phi}-\dot{\theta})\operatorname{sen}(\phi-\theta)\right)$$

$$-\ddot{\ell}\operatorname{sen}(\phi-\theta) - \dot{\ell}(\dot{\phi}-\dot{\theta})\cos(\phi-\theta) + \dot{\ell}\dot{\theta}\cos(\phi-\theta) + \ell\ddot{\theta}\cos(\phi-\theta) - \ell\dot{\theta}(\dot{\phi}-\dot{\theta})\operatorname{sen}(\phi-\theta) = \left(R\dot{\phi}^2 - \dot{\phi}\dot{\ell}\cos(\phi-\theta) - \ell\dot{\phi}\dot{\theta}\operatorname{sen}(\phi-\theta)\right) - mg\operatorname{sen}\phi = Q_r^L$$